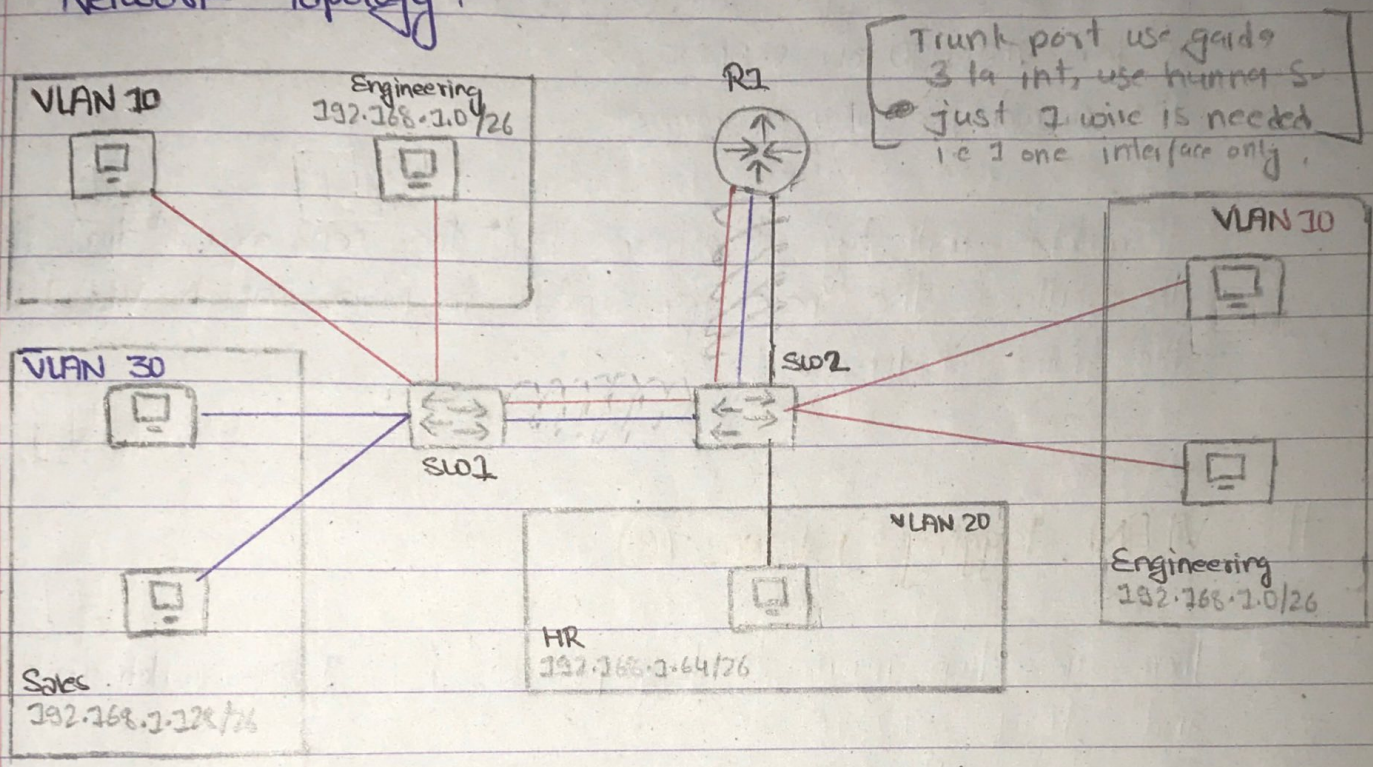
Day-17

Oct 22

VLANS (Part 2).

Network Topology.



Trunk Ports. (Tagged ports).

- In a small network with few VLANS, it is possible to use separate interface for each VLAN when connecting switches to switches, and switches to routers.
- However, when no. of VLANS increases, this isn't viable. It will result in wasted interfaces, and often routers won't have enough interfaces for each VLAN.
- We can use 'trunk ports' to carry traffic from multiple VLANS over a single interface.

After using trunk ports, we don't need separate connection for each VLAN. So there'll be single connection between SW1-SW2 & SW2-Router. These are single physical connection, but traffic from multiple VLANS is allowed over each trunk. Lets see an example.

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SW1 gets a frame (for VLAN 10), but how exactly it knows it is for VLAN 10 or VLAN 30?

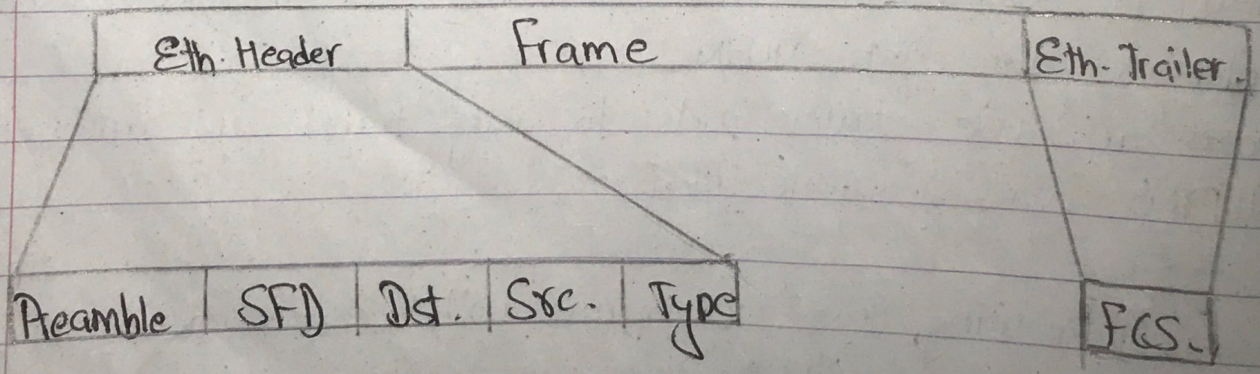
4) The answer is Vlang tagging.

Switches will 'tag' all frames that they send over a trunk link. This allows the receiving switch to know which VLAN the frame belongs to.

VLAN Tagging (ISL/802.1Q)

- There are two main trunking protocols: ISL (Inter-Switch Link and IEEE 802.1Q.
- ISL is an old CISCO proprietary protocol created before the industry standard IEEE 802.1Q.
- IEEE.1Q is an industry standard protocol created by IEEE.
- We'll probably never use ISL in real world. Even moder CISCO equipment doesn't support ISL. We need to learn IEEE@ 802.1Q for CCNA

Ethernet Frame.



802.1Q (a.1Q / dot1Q) tag is inserted between two fields (Src & Type) of Ethernet Header.

* Ethernet Header with 802.1Q Tag.

Preamble	SFD	Destination	Source	802.1Q	Type/Length

- Dot1Q is a 4-byte or 32 bit field.
- The tag (.1Q) consists of two main fields.
  i) Tag Protocol Identifier (TPID)
  ii) Tag Control Information (TCI)

- TCI itself consists of 3 subfields: PCP, DEI, VID.

16 bits	3 bits	1 bit	12 bits.
TPID TPID	TCI		
	PCP	DEI	VID

* TPID
- It is 2 bytes (16 bits) in length.
- It is always sets to value of 0x8100. (0x represents hexadecimal)
So, 8100 = 4 hex chars = 16 bits. This tag indicates that the frame is 802.1Q tagged.
- When switch sees value 8100, it knows it's dot1Q tagged frame.

* TCP & TCI

a) PCP (Priority Code Point)
- Its a 3 bit in length
- It is used for CoS (Class of Service), which prioritizes important traffic in congested networks.

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b) DEI (Drop Eligible Indicator).

- It is single bit in length
- It is used to indicate frame that can be dropped if the network gets congested. (so that more imp. traffic goes through)

c) VID (Vlan Id)

- It is 12 bit in length
- Identifies the VLAN the frame belongs to
- Most important field of .1Q Tag.
- 12 bits in length = 4096 total VLANS (2^{12})
  = range of 0-4095.

- Vlans 0 and 4095 are conserved & can't be used.
- Therefore the actual range of VLANS is 1-4094.
- CISCO's proprietary ISL also has a VLAN range of 1-4094.

VLAN ranges.

The range of VLANS (1-4094) is divided into two sections:

Normal VLANS - 1-1005

Extended VLANS - 1006-4094

Some older devices can't use the extended VLAN range, however it's safe to expect that moder switches will support the extended VLAN range.

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Note: A standard layer 2 switch will only forward traffic in the same VLAN, it won't forward traffic between VLANS.

Native VLAN.

- 802.1Q has a feature called native VLAN. (ISL doesn't have this feat)
- The native VLAN is VLAN 1 by default on all trunk ports, however this can be manually configured on each trunk port.
- The switch doesn't add an 802.1Q tag to frames in native LAN.
- When a switch receives an untagged frame on trunk port, it assumes the frame belogs to native VLAN.

It's very important that native VLAN matches of sending & receiving switch.

Trunk Configuration.
(Lets take SW1 & SW2 in our case)

#
interface g0/0
switchport mode trunk.
(If trunk encapsulation auto नहीं है तो पहले use need to set it to .1Q)
switchport trunk encapsulation ~~dot1Q~~ dot1q
then do switchport mode trunk.

show interfaces trunk
(~~shows~~ lists trunk interfaces). Mode = on setuped manually

Configuring VLANs allowed on trunk.

int g0/0
switchport trunk allowed vlan 10,30
(यो यहाँदि first मा configure गर्दा)
(यसले 10 र 30 vlan configure गर्छ, if we wanna add 20, we d.

switchport trunk allowed vlan add 20
(Now if we wanna remove 20 then)
switchport trunk allowed vlan remove 20
switchport " " vlan all (adds allows all vlan 1-4094)
This is same as default state as in default state too, all vlans are allowed.
Switchport trunk allowed vlan except 1-5,10
switch port " " " none.
(No vlans allowed on trunk)

For security purposes, it is best to change the native VLAN to unused VLAN.

Changing native VLAN.

int g0/0 =/(then switchport mode trunk)
switchport trunk native vlan 1001

§ ROAS is used to route between multiple VLANs using a single interface on router & switch §.

Router on a Stick (ROAS)

- It's a method of inter-VLAN routing, where there is only a single physical interface connecting router & switch.
- In our network G0/0 int. of router is used to connect to SW2. trunk G0/1 interface.
- We can actually divide this one physical interface into three separate ~~sub-int~~ subinterfaces. which'll allows to to perform inter-VLAN routing with only one physical interface.
  So, it'll look like G0/0.10, G0/0.20 & G0/0.30 for VLAN ~~VLAN~~ 10, 20 & 30 respectively.
  These 3 logical sub-interfaces are really one physical interface, G0/0 which is connected to SW2's G0/1, but they can operate like three separate interfaces.
  We don't need to do any additional config on SW2,

```
R1 # int g0/0
R1 # no shutdown
R1 # int g0/0.10   (Sub-int no. vlan no. match na garda ni kei hunxa, but its recommended that they match as it makes job easier)
R1 #
R1 # encapsulation dot1q. 10   (vlan no = 10)
R1 # ip add 192.168.1.62 255.255.255.192
```

Then check using

```
R1# do sh ip int brief
R1 # doshow ip routes   (connected & local routes are added as just as when IP addresses are added to physical int)
```

VLAN

1) You want to configure SW1 to send VLAN 10 frames untagged over its GigabitEthernet 0/1 interface, a trunk. Which command is appropriate.
a) encapsulation dot1q 10
b) switchport trunk allowed vlan 10
c) switchport " " vlan add 10
d) switchport trunk native vlan 10

2) After modifying list of VLANS allowed on trunk int, you want to return to default state whatis the command.
a) switchport trunk allowed vlan default
b) " " allowed vlan all
c) " " allowed vlan, 1,1001-1005

3) You tried to configure int. on a Cisco switch as trunk port with command switchport mode trunk, but command is rej, Which cmnd might fix this.
a) switchport mode trunk
b) switchport trunk encapsulation 802.1q
c) " " " dot1q
d) " " " auto.

4) Which field of 802.1q tag identifis VLAN id of frame)
a) TPID b) VID c) PCP d) VLN

5) U configd a switchport trnk allowd vlan 10 on int, but vlan 10 doesn't appear in vlan allowed & active in management cmd checked using show interfaces trunk. What might be reason
a) VLAN 10 doesn't exist on switch